

# Nusaibah Binte Rawnak

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## EDUCATION

### McGill University

Montreal, QC (January 2023 – April 2026)

- B.A.Sc., Cognitive Science: Computer Science and Linguistics (Minor Statistics)
- **A2-level French** coursework with focus on oral and written communication; currently pursuing **B1**.

## WORK EXPERIENCE

### bKash

Dhaka, Bangladesh (May 2025 - August 2025)

Software Engineering Intern

- Developed and integrated **RESTful APIs** in **Java** and **SQL** to enhance financial data processing pipelines for the bKash mobile banking application, supporting a microservices-based backend architecture.
- Automated test suites using **JUnit** to validate API functionality and improve software reliability, reducing manual testing overhead through systematic debugging and regression coverage.

### City Bank

Dhaka, Bangladesh (June 2024 - August 2024)

Data Science Intern

- Built and deployed a predictive machine learning model with **Scikit-learn** and **PyTorch** to identify high-value customers, boosting deposit-targeting efficiency by **15%**.
- Designed and executed **ETL** pipelines using **Pandas**, **spaCy**, and **NLTK** to extract, clean, and transform structured and unstructured data; applied **Seaborn** visualization to surface behavioral patterns improving customer targeting.

### Marie Stopes

Dhaka, Bangladesh (May 2023 - August 2023)

Business & Data Analytics Intern

- Performed **data mining** on client experience datasets using **Python** and **Excel (VBA)** to uncover engagement trends and service gaps, informing data-driven marketing strategies.
- Developed **interactive Power BI** and **Tableau dashboards** backed by structured reporting layers to visualize insights and guide evidence-based decisions, improving client retention and campaign performance.

## PERSONAL PROJECTS

### Track Split: AI-Powered Expense Tracker – TypeScript, Python

- Currently developing an expense platform with dynamic visualizations and **ML-driven** spending insights
- Integrating **OCR-based** bill scanning and intelligent group-splitting to streamline expense management
- Built with a **Next.js** frontend, serverless API routes, and **PostgreSQL** via **Prisma**, deployed on **Vercel**.

### Context Helps, But Only at Scale: Evaluating Practical LLMs for Time Series Forecasting

- Benchmarked LLM architectures (**Llama 3.2**, **GPT-4o-mini**, **Mixtral**) against classical methods (**ARIMA**, **ETS**) across 120 forecasting tasks spanning 12 domains including finance, energy, and transportation.
- Developed **XGBoost routing classifier** achieving **89.5%** of optimal performance while reducing LLM usage by **69%**, enabling practical deployment for resource-constrained environments.

## LEADERSHIP EXPERIENCE

### Rawnak's Bakery

Dhaka, Bangladesh and Montreal, QC (June 2022 - Present)

Founder and Manager

- **Founded and scaled** a home-based bakery, achieving consistent **revenue growth** through innovative digital marketing and customer engagement strategies.
- Leveraged **sales and operational data** to optimize inventory, streamline workflows, and manage **end-to-end operations** - from product development to financial planning - ensuring high customer satisfaction.

### McGill University

Montreal, QC (September 2025 – April 2026)

Course Assistant, MATH 240 (Discrete Structures) & MATH 323 (Probability)

- Led **weekly tutorials** and graded assignments for ~330 students via **Crowdmark**, across two consecutive terms covering **discrete structures** and **probability theory**.
- Provided **targeted feedback** to strengthen **analytical reasoning** while coordinating with instructors to maintain grading standards and lesson consistency.

## TECHNICAL SKILLS

- **Languages:** Python, R, Java, C, SQL, C++, HTML, CSS, JavaScript, Clojure, LaTeX
- **Tools & Scripting:** Bash, Git, VS Code, Linux/Unix, Figma, Excel (VBA), Power BI, Tableau, spaCy, NLTK, Google Colab, Vercel
- **Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, Prisma, React, Next.js
- **Data Engineering:** ETL Processing, Data Mining, Data Warehousing, Database Development, PostgreSQL
- **Machine Learning & Data Analysis:** Neural Networks, NLP(Transformers, BERT, Prompt Tuning), Time-Series Forecasting, Reinforcement Learning